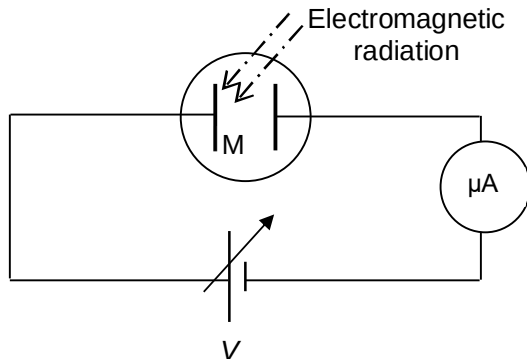


- 26 A 0.31 mW beam of photons with energy 3.11 eV each is incident on a clean metal plate M. The potential difference V is varied until the microammeter gives a maximum reading of $2.0 \mu\text{A}$.



What is the ratio $\frac{\text{electrons emitted per unit time}}{\text{photons incident per unit time}}$?

A 3.2×10^{-21}



B 2.0×10^{-2}

C 1.0

D 1.3×10^{17}

Electrons: $dN_e/dt = I / e = 2.0 \times 10^{-6} / 1.6 \times 10^{-19}$

Photons: $dN_p/dt = P / E = 0.31 \times 10^{-3} / 3.11 \times 1.6 \times 10^{-19}$

- 27 What is the de Broglie wavelength of an alpha particle of kinetic energy E , given that m is the